

Table 1 Variables, sets, and parameters of the TNLAP formulation

<i>Sets</i>	
A	Set of articles: $A = \{1, \dots, n\}$
$A_{j,l,k}$	Set of articles that can be placed on page j , master layout l , box k due to editorial section constraints: $A_{j,l,k} \subseteq A$
P	Set of pages: $P = \{1, \dots, m\}$
M	Set of master layouts: $M = \{1, \dots, t\}$
M_j	Set of available master layouts based on page j : $M_j \subseteq M$
B_l	Set of boxes available in master layout l : $B_l = \{1, \dots, b_l\}$
S	Set of shells: $S = \{1, \dots, r\}$
S_i	Set of shells available for article i , depending on required media elements (e.g., images): $S_i \subseteq S$
$S_{l,k}$	Set of shells available in master layout l , box k : $S_{l,k} \subseteq S$
$S_{i,l,k}$	Set of shells available to both article i and box k in master layout l : $S_{i,l,k} = S_i \cap S_{l,k}$
<i>Parameters</i>	
n	Number of articles
m	Number of pages
t	Number of master layouts
b_l	Number of boxes in master layout l
r	Number of shells
l_i	Length of article i
q_i	Editorial priority of article i , beispiel 1, 0.5, 0, 1
α	Weighting factor, where $0 \leq \alpha \leq 1$
$\min_s, \max_s$	Minimum and maximum character count of shell s
T^u, T^o	Relative thresholds for under- and overfilling
G_{fix}^u	Fixed penalty for underfilling, $G_{\text{fix}}^u = 0.51$
G_{var}^u	Variable penalty for underfilling, $G_{\text{var}}^u = 0.48$
G_{fix}^o	Fixed penalty for overfilling, $G_{\text{fix}}^o = 0.01$
G_{var}^o	Variable penalty for overfilling, $G_{\text{var}}^o = 0.49$
G^e	Penalty if a box is considered empty, $G^e = 1$
C	Sufficiently large number
<i>Variables</i>	
$x_{i,j,l,k,s} \in \{0, 1\}$	Indicator, whether article i is placed on page j master layout l , box k , shell s
$y_{j,l} \in \{0, 1\}$	Indicator, whether master layout l is selected on page j
$z_{j,l,k,s} \in \{0, 1\}$	Indicator, whether shell s is activated on page j , master layout l , box k
$v_{j,l,k} \in \{0, 1\}$	Indicator, whether box k , on page j , master layout l is empty
$\delta_{j,l,k}^u \in \{0, 1\}$	Indicator, whether box k , on page j , master layout l is underfilled
$\delta_{j,l,k}^o \in \{0, 1\}$	Indicator, whether box k , on page j , master layout l is overfilled
$e_{j,l,k}^u \in \{0, 1\}$	Indicator, whether the underfilling of box k on page j , master layout l exceeds the threshold
$e_{j,l,k}^o \in \{0, 1\}$	Indicator, whether the overfilling of box k on page j , master layout l exceeds the threshold
$\min_{j,l,k}^{\text{box}} \in \mathbb{R}_0^+$	Minimum character count of box k on page j , master layout l
$\max_{j,l,k}^{\text{box}} \in \mathbb{R}_0^+$	Maximum character count of box k on page j , master layout l
$p_{j,l,k} \in \mathbb{R}_0^+$	Priority of the article placed in box k on page j , master layout l
$c_{j,l,k} \in \mathbb{R}_0^+$	Penalty term for box k on page j , master layout l
$c_{j,l,k}^u \in \mathbb{R}_0^+$	Penalty term for underfilling in box k on page j , master layout l
$c_{j,l,k}^o \in \mathbb{R}_0^+$	Penalty term for overfilling in box k on page j , master layout l
$f_{j,l,k} \in \mathbb{R}_0^+$	Fitness of box k on page j , master layout l
$f_j \in \mathbb{R}_0^+$	Fitness of page j

$$\max \sum_{j \in P} f_j \quad (1)$$

$$\text{s.t.} \sum_{l \in M_j} y_{j,l} = 1 \quad \forall j \in P \quad (2)$$

$$\sum_{s \in S_{l,k}} z_{j,l,k,s} = y_{j,l} \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (3)$$

$$x_{i,j,l,k,s} \leq z_{j,l,k,s} \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l, \forall i \in A_{j,l,k}, \forall s \in S_{i,l,k} \quad (4)$$

$$\sum_{j \in P} \sum_{l \in M_j} \sum_{k \in B_l} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \leq 1 \quad \forall i \in A \quad (5)$$

$$\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} = y_{j,l} - v_{j,l,k} \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (6)$$

$$\sum_{s \in S_{l,k}} z_{j,l,k,s} \cdot \min_s = \min_{j,l,k}^{\text{box}} \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (7)$$

$$\sum_{s \in S_{l,k}} z_{j,l,k,s} \cdot \max_s = \max_{j,l,k}^{\text{box}} \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (8)$$

$$\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot l_i \geq \min_{j,l,k}^{\text{box}} - C \cdot (\delta_{j,l,k}^u + v_{j,l,k}) \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (9a)$$

$$\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot l_i \leq \min_{j,l,k}^{\text{box}} + C \cdot (1 - \delta_{j,l,k}^u) \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (9b)$$

$$\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot l_i \leq \max_{j,l,k}^{\text{box}} + C \cdot \delta_{j,l,k}^o \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (10a)$$

$$\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot l_i \geq \max_{j,l,k}^{\text{box}} - C \cdot (1 - \delta_{j,l,k}^o) \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (10b)$$

$$\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot l_i \leq (1 - T^u) \cdot \min_{j,l,k}^{\text{box}} + C \cdot (1 - e_{j,l,k}^u) \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (11a)$$

$$\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot l_i \geq (1 - T^u) \cdot \min_{j,l,k}^{\text{box}} - C \cdot (e_{j,l,k}^u + v_{j,l,k}) \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (11b)$$

$$\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot l_i \leq (1 + T^o) \cdot \max_{j,l,k}^{\text{box}} + C \cdot e_{j,l,k}^o \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (12a)$$

$$\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot l_i \geq (1 + T^o) \cdot \max_{j,l,k}^{\text{box}} - C \cdot (1 - e_{j,l,k}^o) \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (12b)$$

$$p_{j,l,k} \leq \sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot q_i \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (13)$$

$$c_{j,l,k}^u \geq G_{fix}^u - \left(\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot \frac{l_i}{\min_s} - 1 \right) \cdot \frac{G_{var}^u}{T^u} - (1 - \delta_{j,l,k}^u + e_{j,l,k}^u) \cdot C \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (14)$$

$$c_{j,l,k}^o \geq G_{fix}^o + \left(\sum_{i \in A_{j,l,k}} \sum_{s \in S_{i,l,k}} x_{i,j,l,k,s} \cdot \frac{l_i}{\max_s} - 1 \right) \cdot \frac{G_{var}^o}{T^o} - (1 - \delta_{j,l,k}^o + e_{j,l,k}^o) \cdot C \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (15)$$

$$c_{j,l,k} = y_{j,l} - \left(c_{j,l,k}^u + c_{j,l,k}^o + (e_{j,l,k}^u + e_{j,l,k}^o + v_{j,l,k}) \cdot G^e \right) \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (16)$$

$$f_{j,l,k} = \alpha \cdot p_{j,l,k} + (1 - \alpha) \cdot c_{j,l,k} \quad \forall j \in P, \forall l \in M_j, \forall k \in B_l \quad (17)$$

$$f_j = \sum_{l \in M_j} \frac{1}{|B_l|} \sum_{k \in B_l} f_{j,l,k} \quad \forall j \in P \quad (18)$$